Optimization for Deep Learning (OPT4DL)

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Momentum as Bias-Variance Tradeoff

Momentum and Gradient Estimation



- Recall under the SFO assumption, g_t is (conditionally) an unbiased estimator of $\nabla f(w_t)$ with variance σ^2
- In SGDm, we use $m_t = \beta m_{t-1} + (1-\beta) g_t$, $m_0 = 0$ instead of g_t to update our parameter
- Note that from an estimation perspective, this leads to bias as

$$\begin{aligned} \mathbb{E}_{z_t}[m_t | z_{1:t-1}] &= \mathbb{E}_{z_t}[\beta m_{t-1} + (1-\beta) g_t | z_{1:t-1}] \\ &= \beta \mathbb{E}_{z_t}[m_{t-1} | z_{1:t-1}] + (1-\beta) \mathbb{E}_{z_t}[g_t | z_{1:t-1}] \\ &= \beta \mathbb{E}_{z_t}[m_{t-1} | z_{1:t-1}] + (1-\beta) \nabla f(w_t) \neq \nabla f(w_t) \end{aligned} \tag{1}$$

- What is the impact on convergence? Let us assuming L-smoothness

$$f(w_{t+1}) \leq f(w_t) - \eta \langle m_t, \nabla f(w_t) \rangle + \frac{L\eta^2}{2} \|m_t\|^2 \tag{2}$$

The usual analysis breaks since the $\mathbb{E}[\langle m_t, \nabla f(w_t) \rangle | z_{1:t-1}] \neq \|\nabla f(w_t)\|^2$.

Role of Estimation Error



- Let $e_t := m_t - \nabla f(w_t)$ be our total estimation error for SGDm
- Recall for SGD $e_t := g_t - \nabla f(w_t)$ such that $\mathbb{E}[\|e_t\|^2 | z_{1:t-1}] \leq \sigma^2$

Lemma 1

If $\eta \leq 1/4L$, for SGDm we have

$$\|\nabla f(w_t)\|^2 \leq \frac{4}{\eta}\Big(f(w_t) - \mathbb{E}_{z_t}[f(w_{t+1})|z_{1:t-1}]\Big) + 3\mathbb{E}[\|e_t\|^2|z_{1:t-1}] \tag{3}$$

- Recall, we had a result for SGD the second term was effectively $\eta\sigma^2$
- Thus, if we hope for convergence, $\mathbb{E}[\|e_t\|^2|z_{1:t-1}]$ must be small, e.g. $\mathcal{O}(\eta)$. That is, despite the small bias, variance reduces by a lot, hence estimation error and in turn convergence error reduces.

Evolution of the Error

- Let us rewrite Let $e_{t+1} := m_{t+1} - \nabla f(w_{t+1})$

$$\begin{aligned} e_{t+1} &= \beta m_t + (1-\beta) g_{t+1} - \nabla f(w_{t+1}) \pm \nabla f(w_t) \\ &= \beta e_t + (1-\beta)(g_{t+1} - \nabla f(w_{t+1})) + \beta(\nabla f(w_t) - \nabla f(w_{t+1})) \end{aligned} \tag{4}$$

- When $\beta < 1$ the first term vanishes exponentially. The second term is also small and assuming a slow evolution of model parameters $\nabla f(w_t) \approx \nabla f(w_{t+1})$ and the third term is also small.

Lemma 2

Under the SFO assumption, SGDm's error satisfies the following recursion

$$\mathbb{E}\|e_{t+1}\|^2 \leq \beta \mathbb{E}\|e_t\|^2 + A(1-\beta)\mathbb{E}\|\nabla f(w_t)\|^2 + \tilde{A}(1-\beta)^2\sigma^2 \tag{5}$$

for some positive constants $A, \tilde{A} > 0$ when $\eta = C(1-\beta)/L$.

Convergence Result



Theorem

Assume $f(w) = \mathbb{E}[\ell(w, z)]$ is L-smooth. Furthermore, assume we have access to an SFO and that $z_1, \ldots, z_T$ are statistically independent. Then, SGDm with learning rate $\eta = C(1-\beta)/L$ and momentum parameter $1 - \beta = \mathcal{O}(1/\sqrt{T})$ satisfies $\mathbb{E}\|\nabla f(\hat{w})\|^2 \simeq \mathcal{O}(1/\sqrt{T})$.

- The theorem highlights that the momentum parameter should be very large, e.g., $\beta = 0.99$, consistent with practice.
- Similar results hold under normalization

$$m_t = \beta m_{t-1} + (1-\beta) g_t, \qquad w_{t+1} = w_t - \eta \frac{m_t}{\|m_t\| + \kappa} \tag{6}$$

- Thus, despite a biased estimation, convergence occurs as the estimation error vanishes, thanks to variance reduction, as the algorithm converges (a virtuous cycle). Other/better ways to reduce variance?

Proof Sketch



- The proof relies on the idea of defining a potential function and tracking its evolution
- We actually had potential functions before which we worked with implicitly:

$$\Phi_t = \|w_t - w^*\|^2 \quad \text{for cvx and Lip. function,} \qquad \Phi_t = \mathbb{E}[f(w_t)] - f^* \quad \text{for smooth function} \tag{7}$$

  and we studied $\Phi_{t+1} - \Phi_t = \mathbb{E}[f(w_{t+1}) - f(w_t)]$ using L-smoothness
- Generalizable Template: Augment with above sources of errors, in this case, the estimation error

$$\Phi_t = \mathbb{E}[f(w_t)] - f^* + C_e \mathbb{E}\|e_t\|^2 \tag{8}$$

- Study the evolution of potential by summing over its consecutive differences:

$$\sum_{t=1}^{T} \Phi_{t+1} - \Phi_t = \Phi_{T+1} - \Phi_1 \geq f^* - f(w_1) - C_e \mathbb{E}\|e_1\|^2 \tag{9}$$



- By the definition of potential

$$\sum_{t=1}^{T} \Phi_{t+1} - \Phi_t = \left(\sum_{t=1}^{T} \mathbb{E}[f(w_{t+1}) - f(w_t)] \right) + \left(C_e \sum_{t=1}^{T} \mathbb{E}\|e_t\|^2 \right) \tag{10}$$

- First term dealt with by Lemma 1 (Smoothness) and Second term dealt with by Lemma 2 (evolution of estimation error)
- The nice property of error

$$\mathbb{E}\|e_{t+1}\|^2 \leq \beta \mathbb{E}\|e_t\|^2 + \text{ other terms}, \qquad 0 < \beta < 1 \tag{11}$$

ensures we can find a good constant C_e to ensure convergence.

- This idea can be utilized in a variety of learning settings (we will discuss a few more cases).